

Programming assignments#3

Climb Smart: Minimizing Stair Costs

1. Description

You are climbing a staircase with n levels. From level 0 to level $n-1$, each step has a different climbing cost. The climbing cost means that when you step on a particular level, you need to pay the corresponding cost.

You will receive an integer array `cost`, where `cost[i]` represents the climbing cost of the i -th level. After paying this cost, you can choose to climb either 1 level or 3 levels. At the beginning, you can choose to start from either level 0 or level 1.

Please write a function to calculate the minimum cost to reach the top of the staircase.

Note that reaching the top level can be done by stopping exactly at the top, or by exceeding it by any number of steps.

Example1:

Input: `cost = [10, 15, 20]`

Output: 10

Explanation: You will start at index 0.

- Pay 10 and climb 3 steps to reach the top (index 3).
- So The total cost is 10.

Example2:

Input: `cost = [1, 100, 1, 1, 1, 100, 1, 1, 100, 1]`

Output: 4

Explanation: You will start at index 0.

- Pay 1 and climb 3 steps to reach index 3.
- Pay 1 and climb 3 steps to reach index 6.
- Pay 1 and climb 3 step to reach index 9.
- Pay 1 and climb 1 step to reach the top (index 10).
- So The total cost is 4.

Example3:

Input: cost = [5, 2, 3, 10, 1, 6]

Output: 3

Explanation: You can start at index 1.

- Pay 2 and climb 3 steps to reach index 4.
- Pay 1 and climb 3 step to reach the top.
- So The total cost is 3.

2. Input Format

The input is an integer array called cost.

10 15 20

Taking Example 1 as an example, the array has 3 elements:

- The first element, 10, represents the cost of climbing to level 0.
- The second element, 15, represents the cost of climbing to level 1.
- The third element, 20, represents the cost of climbing to level 2.

Since there are 3 elements in the array, this means that the top level is level 3.

Input Constraints:

$2 \leq \text{cost.length} \leq 1000$

$0 \leq \text{cost}[i] \leq 999$

3. Output Format

Print the minimum cost to reach the top of the staircase.

10

4. Program

1. Program in C/C++.

5. Grading

1. Score:
 - ♦ Q1~Q10: Each contributes 10%
 - ♦ Q1~Q5 are open case
 - ♦ Q6~Q10 are hidden case
2. The score of questions is evaluated by the OJ system. **TA will evaluate your grades based on the most recent version of your submissions.**

6. E3 Submission

1. Submit your source file to the E3 system
[Student_ID_Number]_hw3.cpp (.c)
2. Please submit the source code of your latest submission on the OJ. Please make sure that all characters of the filename are in **lowercase**. For example, if your student number is 313510224, the name of your source file should be "313510224_hw1.cpp".
3. Remember the submission rules mentioned above, or you will be penalized by **-15 points** on your grade.

7. Policy

1. The upload deadline would be at **23:59 on November 20, 2024**. The score will be **30% off** if your submission is late.
2. **Do not plagiarize. If you plagiarize, you will receive a score of zero.**

8. Problems

1. If you have any problem with this project, please post it on the E3 forum.
2. If you have any private problem, you can send emails to:
yulin.ee13@nycu.edu.tw